

Bounded Possibility

Constraint, Recurrence, and the Emergence of Meaningful Probability

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Exploratory working paper. This document develops a conceptual hypothesis and a possible formal vocabulary. It does not claim that the proposed quantities are already validated physical observables.

Abstract

This paper develops the hypothesis that physical constraint does more than reduce freedom. A boundary can preserve return paths, retain the consequences of transformation, and keep otherwise dispersive possibilities inside one continuing relational account. When energy opens a wide field of reachable states while confinement prevents those states from dissolving out of the system, possibility and probability are not opposites. Possibility describes the transformations made reachable; probability describes how those transformations remain accountable to the bounded whole. The proposed thesis is that meaningful probability emerges when possibility is densely connected, repeatedly revisited, and retained long enough for stable weighting to form. In this view, an open system may contain many microscopic interactions while losing the relations that make those interactions belong to the same body. A constrained energetic system can support more recursive relationships and therefore more recoverable state transitions. Meaning is treated here in an operational sense: a difference has meaning when it survives transformation sufficiently to influence, return, and be distinguished again. The paper then applies this framework to phase change, plasma, particle isolation, and quantum measurement, while clearly separating the conceptual proposal from established thermodynamics and quantum theory.

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1. The Starting Intuition

Consider water heated in an open saucepan. Liquid becomes vapour, the vapour enters the surrounding air, and its subsequent interactions disperse through a larger environment. Now consider the same transformation inside a sealed or strongly confining vessel. The material cannot simply leave. Pressure changes, vapour remains available to condense, heat can be added and removed, and the same material can repeatedly traverse different phases.

Open: liquid $\rightarrow$ vapour $\rightarrow$ air $\rightarrow$ dispersal

Bounded: liquid $\rightleftharpoons$ vapour $\rightleftharpoons$ wall $\rightleftharpoons$ pressure $\rightleftharpoons$ liquid

The difference is not merely that one system has a wall and the other does not. The boundary changes which relations can return. It keeps the products of a transformation available to participate in later transformations. The system becomes capable of changing state repeatedly rather than producing a one-way local loss.

The container does not merely hold matter. It makes the transitions of that matter continue to belong to one body.

2. Open Dispersal Is Not the Absence of Interaction

Steam released into air does not stop interacting. It interacts with air molecules, temperature gradients, surfaces, humidity, and atmospheric motion. The important loss is not the number of microscopic relationships. The important loss is that those relationships cease to be recursively accountable to the original local system.

many interactions $\neq$ one retained relational history

An open system can therefore generate enormous relational activity while losing the ability to recover its own earlier states. The correlations are exported. The local system no longer has access to the same material, energy, and consequences required to close the cycle.

Open dispersal	Bounded recurrence
Relations spread into an enlarging environment.	Relations repeatedly return through the same boundary conditions.
Material and energy may leave the local account.	Material and energy remain more available to the local account.
Many paths become exits.	Many paths become loops or revisitable transitions.
The original body loses recoverability.	The body retains a shared history across state change.

3. The Container as a Relational Operator

A container should not be treated merely as another object placed around the system. Functionally, it is an operator on the system's transition structure. It suppresses some paths, redirects others, and creates returns that would not exist in the same form without confinement.

$B : \text{transition network} \rightarrow \text{retained transition network}$

The boundary may reduce direct escape while increasing recurrence. In doing so, it changes not only where the system is located but what the system can repeatedly become. It can preserve phase cycling, pressure feedback, collision frequency, and delayed return. Constraint therefore changes the topology of possibility.

A boundary removes dispersive paths so that transformative paths can remain available.

4. Constraint Can Create Physical Possibility

In ordinary speech, possibility is often imagined as whatever could happen in the absence of restriction. Physical possibility is more demanding. A state is not physically possible for a particular system merely because it can be imagined. There must be an accessible path from the present condition to that state, and the resources required for the transition must remain available long enough for it to occur.

$\text{physical possibility} = \text{allowed} + \text{reachable} + \text{retainable}$

By preventing vapour from escaping, a vessel removes the path of immediate dispersal but preserves the paths of condensation, renewed evaporation, pressure equilibration, and repeated heat exchange. Constraint has removed one kind of freedom while creating a larger reusable family of local transformations.

This gives a direct reversal of the usual assumption: less freedom of escape can produce more freedom of transformation. Constraint does not necessarily make the possibility-space smaller. It may make the possibility-space more connected and more reusable.

5. The Tightest Leash on Probability and the Longest Leash on Possibility

A highly energetic, highly constrained system combines two operations that appear opposite only when they are placed on the same axis. Energy opens access to more transformations. Constraint requires those transformations to remain accountable to the same system.

$\text{energy} \rightarrow \text{breadth of reachable possibility}$

$\text{constraint} \rightarrow \text{continuity of probabilistic accountability}$

This is the sense in which a system can place the tightest leash on probability while placing the longest leash on possibility. The system is free to explore many states, but it is not free to pretend that each state belongs to a different universe of account. Conservation, boundary conditions, and prior interactions continue to weight what can actually occur.

The richest production may occur when possibility is opened energetically while probability is forced to keep resolving inside one conserved relational whole.

6. Dense Possibility

The phrase dense possibility does not mean an arbitrary number of imagined outcomes. It means a high concentration of reachable states connected by many available transitions, especially transitions that can return, recur, or feed into one another.

Let the possible system states form a directed graph $G = (V, E)$. The vertices V are distinguishable states and the edges E are allowed transitions. An open dispersing system may contain extensive branching while having few return paths. A confined system can convert branches into cycles.

Dispersive: $S \rightarrow A \rightarrow A_1 \rightarrow \text{outward loss}$

Recurrent: $S \rightarrow A \rightarrow B \rightarrow S$

Dense possibility is therefore not only a large number of states. It is a large number of mutually available transformations. Possibilities become structurally meaningful when reaching one state does not automatically destroy access to the rest of the state-space.

possibility density $\approx$ reachable states $\times$ transition connectivity $\times$ recurrence

The final expression is a heuristic definition, not yet a physical law. Its purpose is to expose the three ingredients that the ordinary word possibility tends to collapse together.

7. Meaningful Probability

Probability becomes meaningful when the system persists long enough for its possibilities to be tested repeatedly under a continuing set of constraints. If the first transition destroys or exports the system, the alternatives function mainly as possible exits. When paths can be revisited, the system can establish stable relative weighting.

$$S \rightleftharpoons A, \quad S \rightleftharpoons B, \quad S \rightleftharpoons C, \quad S \rightleftharpoons D$$

$$p(A) + p(B) + p(C) + p(D) = 1$$

The normalisation condition is not itself the new idea. The proposed interpretation is that a probability distribution becomes physically informative because its alternatives remain members of one retained account. Probability is not simply painted onto abstract possibility. It is the pattern that emerges when reachable possibilities repeatedly encounter the same body, invariants, and boundary conditions.

Meaningful probability is possibility repeatedly tested inside persistent constraint.

8. Meaning as Recoverable Difference

Meaning is used here in an operational rather than psychological sense. A distinction has physical meaning when it survives long enough, or leaves enough recoverable consequence, to affect a later distinction. A difference that is immediately erased without trace cannot participate in a continuing structure.

difference $\rightarrow$ interaction $\rightarrow$ retained consequence $\rightarrow$ possible return

difference $\rightarrow$ immediate dispersal $\rightarrow$ unrecoverable loss

The first sequence can accumulate history. The second cannot easily be distinguished from noise from the standpoint of the original system. Meaning therefore requires neither permanence nor a fixed object. It requires recoverability across transformation.

This connects directly to the semantic traversal system. Persistence does not mean that a thing remains unchanged. It means that enough relation survives the change for the distinction to be reconstructed, answered, or closed.

9. The Body Before the Particle

The same reasoning changes how an isolated particle is interpreted inside a dense plasma. The particle may retain invariant labels such as charge, mass, and spin when analytically separated, but its current behaviour belongs to a larger relational body. Density, temperature, surrounding fields, and repeated collisions determine which local modes are accessible and how they are weighted.

particle alone $\neq$ particle-in-body

Removing a particle from the plasma can produce a cleaner object for measurement, but the act of removal also produces a new system. It eliminates the recurrence, pressure, collective fields, and state transitions that constituted the particle's role inside the plasma.

Isolation may not reveal the object beneath the relations. Isolation may manufacture a new object by removing the relations that previously constituted it.

The body may therefore be more primitive than the particle for some questions. A particle can remain real as a locally recoverable resolution without being the complete carrier of the field's behaviour. A droplet is real, but it does not contain the current of the river merely because both are water.

10. Implications for Measurement

Quantum measurement is commonly narrated as a probe inspecting an object. In physical operation, the probe, target, apparatus, and environment transform one another. The record belongs to the closed chain of interaction rather than to the target alone.

probe $\rightarrow$ medium $\rightarrow$ interaction $\rightarrow$ redistribution $\rightarrow$ record

The present proposal adds a particular emphasis: the measuring relation should not be treated merely as a tool that reveals an already complete object. The relation may be the process through which probe, target, and outcome become separately recoverable identities.

Inside a dense body, attempting to isolate one particle may increasingly measure the boundary conditions of the isolation rather than an independent truth the particle carried before extraction. The useful question may shift from "What does this particle possess?" to "Which distinction remains recoverable across this complete interaction?"

11. A Provisional Formal Scaffold

The following definitions are deliberately provisional. They are offered to organise the idea before any claim of physical derivation.

Term	Provisional operational definition
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Term	Provisional operational definition
Boundary B	An operation that suppresses, redirects, or returns transitions so that consequences remain within a continuing system.
Possibility space ΩB	The set of states reachable from the current state under the constraints and resources preserved by B.
Recurrence R	The degree to which transitions return to, revisit, or remain causally available to the same system.
Distinguishability D	The degree to which alternative states retain differences that can later be recovered.
Probability p	A normalised weighting over reachable alternatives within one retained relational account.
Meaning M	Recoverable difference accumulated across a traversal; not a subjective interpretation.
Object O	A stable equivalence class of recurrent traversals that can be identified across change.

A heuristic meaning-capacity expression may then be written as:

$$M_h \propto H(\Omega B) \cdot R \cdot D$$

Here $H(\Omega B)$ represents the breadth of accessible possibility, R represents recurrence or retained return, and D represents distinguishability. The expression is not yet an empirical equation. It merely states the structural hypothesis that meaning collapses if possibility is absent, if nothing returns, or if all alternatives become indistinguishable.

A second useful quantity would compare recurrent transition weight with outward loss:

$$\rho = \text{returned or retained transition weight} / \text{total transition weight}$$

A system with high energetic access but very low ρ may explore many states while rapidly losing them from the local account. A system with high ρ but almost no energetic access may retain itself while exploring little. The proposed productive regime requires both accessible possibility and sufficient relational retention.

12. What Is Established and What Is Hypothesised

The water example relies on established thermodynamic facts: pressure, temperature, volume, heat transfer, phase equilibrium, and material exchange affect whether water remains liquid, becomes vapour, condenses, or leaves a defined system. Statistical mechanics already treats probabilities as constrained distributions over accessible microstates. Open-system physics already distinguishes local subsystems from their environments.

The new claim is interpretive and structural. It proposes that constraint should be understood not only as limiting a pre-existing possibility-space but also as preserving and connecting the transition paths through which physical possibilities become repeatedly available. It further proposes that probability acquires meaning through retained recurrence inside a body, and that objecthood may arise as a stable resolution of that relational traversal.

Established physical background	Exploratory hypothesis
Boundaries alter pressure, exchange, phase behaviour, and accessible states.	Boundaries can be interpreted as creators of relational return paths.
Higher energy can make additional states or reactions accessible.	Energy lengthens the leash of possibility while constraint tightens probabilistic accountability.
Open systems exchange matter, energy, and correlations with environments.	Rapid dispersal prevents local possibility from accumulating recoverable meaning.
Many-body systems require relational and statistical descriptions.	The body may be ontologically prior to particle-like local resolutions for some descriptions.
Measurement is a physical interaction.	Probe, target, and record may emerge as identities only through closure of the full traversal.

13. Research Questions

1. Can recurrence or relational retention be defined using existing information-theoretic quantities rather than introducing a purely verbal measure?
2. Can the transition network of an open and a confined system be compared through return probability, mixing time, mutual information, or correlation lifetime?
3. Can meaningful probability be distinguished mathematically from a one-pass branching distribution by requiring repeated accessibility or ergodic sampling?
4. Can objecthood be modelled as an equivalence class of traversals whose recoverable invariants survive changes in environment?
5. Can a quantum measurement model be rewritten without installing probe, target, and detector as fully separate objects at the entrance?
6. Where does increased confinement enrich the accessible relational network, and where does it instead erase distinctions through thermalisation or overconstraint?
7. Can the semantic sequence Persist–Distinguish–Relate–Propagate–Resolve–Recurse–Close be mapped onto an explicit dynamical system without forcing an object into the initial state?

14. Conclusion

The central claim is not that constraint is always productive or that maximum density automatically creates meaning. A system can be too rigid to transform, too violent to preserve distinction, or too open to retain consequences. The productive condition lies in the simultaneous presence of accessible alternatives and a boundary capable of keeping those alternatives inside one continuing relational history.

$$\text{meaningful probability} = \text{possibility} + \text{reachability} + \text{recurrence} + \text{accountability}$$

In an open saucepan, vapour continues to interact, but its relations rapidly migrate into an enlarging environment. In a bounded vessel, the same material can repeatedly participate in phase change, pressure response, condensation, and renewed evaporation. The container has not merely reduced freedom. It has made more of the system's transformations mutually available.

This produces the deepest formulation of the idea: possibility does not become meaningful merely by being numerous. It becomes meaningful when many possible transformations are densely connected, retained long enough to encounter one another, and repeatedly resolved inside one body.

Constraint gathers possibility into a shared history. Probability is the meaning that history acquires through repeated resolution.